

SCENARIO 1

Always on the Move

SCENARIO AT A GLANCE

Aleksandra, 37, travels frequently and spends a significant part of the year away from home. Her apartment often continues to consume energy even when it is empty due to heating, standby devices and appliances operating according to regular settings.

What you need to do

- build a realistic historical hourly energy consumption profile for a minimum of 30 consecutive days,
- forecast energy demand for the next 24 hours, with the possibility of extending the forecast to 3 and 7 days,
- use real historical and forecast weather data,
- identify peak demand periods and explain changes in consumption,
- present the results in an application,
- evaluate how renewable energy and better timing of appliance use could reduce grid electricity consumption and household costs.

What you need

You will work with:

- the household profile and description of the resident's behaviour,
- indicative energy consumption parameters for household appliances,
- an example household location for obtaining real historical and forecast weather data,
- your own clearly documented assumptions where additional information is required.

Context

Aleksandra is 37 and travels frequently for both work and personal reasons. Over the course of a year, she spends almost half of her time away from home, although her trips are usually short and irregular rather than long and predictable.

Before leaving, she is often in a hurry and does not always remember to adjust the settings of all household devices. As a result, some appliances remain in standby mode, while the heating system may continue to operate according to its regular schedule even when the apartment is empty.

Aleksandra wants a simple solution that will help her forecast future energy demand, understand when and why consumption will be highest, and identify opportunities to reduce unnecessary energy use without compromising comfort.

Household Profile

1 person	premium city apartment
frequent business and private travel	highly variable occupancy
periods of normal household activity alternating with several hours or several days of absence	electric heating or a heat pump
relatively intensive use of electronic devices	irregular weekly routine

Appliances and Indicative Energy Consumption Parameters

APPLIANCE / SYSTEM	INDICATIVE ENERGY CONSUMPTION
Refrigerator	0.8–1.2 kWh/day
Electric heating / heat pump	1.0–3.0 kW during operation
Kettle	2.0 kW, typically 3–5 min/use
Coffee machine	1.0–1.5 kW, typically 5–10 min/use
Oven	2.0–2.5 kW during operation
Washing machine	0.6–1.0 kWh/cycle
Dishwasher	0.8–1.2 kWh/cycle
Television	0.08–0.15 kW during operation
Laptop	0.04–0.08 kW during operation
Wi-Fi router	0.008–0.015 kW, continuous operation
Lighting	0.05–0.15 kW total when active
Phone / tablet charging	0.005–0.02 kWh/device per charge
Standby devices	0.02–0.06 kW combined



EXAMPLE HOUSEHOLD LOCATION

Warsaw, Poland

The values provided above are indicative and are intended to serve as input parameters for the challenge. Participants may make reasonable additional assumptions where necessary, provided that these assumptions are clearly documented in the final solution.

Participants should use real historical and forecast weather data for the indicated location.

Challenge

Your task is to create a solution that forecasts the household's energy demand and optimises that demand in terms of cost.

1 Historical Energy Consumption Profile

Based on the provided scenario, prepare a realistic historical electricity consumption profile for the household.

The historical profile should cover **a minimum of 30 consecutive days with hourly resolution**.

The profile should:

- reflect the resident's daily routine,
- include appliance operation,
- account for the impact of outdoor temperature,
- include realistic differences between individual days,
- reflect periods when the resident is at home and away,
- reflect the behavioural variability described in the scenario.

The historical profile will serve as input data for the next part of the task.

2 Energy Consumption Forecast

Prepare a model that forecasts the household's hourly energy demand.

The solution must generate a forecast for:

- the next 24 hours.

The solution should also support forecasts for:

- the next 3 days,
- the next 7 days,

depending on data availability.

The forecast should account for predicted weather conditions and the expected presence or absence of the resident.

3 Application

Prepare an application that presents the system's results.

The application must display:

- forecast hourly energy consumption,
- forecast total energy consumption for the selected period,
- outdoor temperature forecast,
- hours with the highest expected demand,
- a brief explanation of the expected increase or decrease in consumption.

The user should be able to view a forecast for at least the next 24 hours.

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Renewable Energy Simulator

The solution should include a module for assessing the impact of an investment in renewable energy sources on Aleksandra's household.

The module should allow the user to select the installation capacity in kWp.

For the purpose of the challenge, it may be assumed that Aleksandra has access to a shared rooftop or community-based photovoltaic installation.

For the selected option, the application should present:

01

estimated annual energy production,

02

**the share of self-generated energy
in covering household demand,**

03

**estimated reduction in electricity
purchased from the grid,**

04

estimated savings,

05

indicative payback period for:

A

a renewable energy installation
without changing current habits,

B

a renewable energy installation
combined with shifting selected
energy-consuming activities
to more favourable periods.

The module should enable comparison of at least several installation capacity options.

We provide common assumptions for electricity tariffs and PV costs for all teams in the shared attachment. For any other assumptions, such as renewable energy production, you are free to choose your own values. Just state them clearly in your solution.

Description of Resident's Behaviour

6:30–9:00	When Aleksandra is at home, she usually starts her day between 6:30 and 7:00. During this time, she: – uses lighting, – uses the kettle and coffee machine, – prepares breakfast, – checks and charges electronic devices, – may use the washing machine before a trip, – expects a comfortable indoor temperature.
9:00–17:00	Her behaviour during this period varies considerably. Depending on the day, Aleksandra may: – work outside the home, – work from home, – prepare for a business trip, – already be travelling. <i>When she is away, the apartment may remain empty for many hours.</i>
17:00–22:30	When at home, Aleksandra: – prepares dinner, – watches television or uses a laptop, – may run the dishwasher, – may use the washing machine, – charges electronic devices, – maintains a comfortable indoor temperature.
22:30–6:30	At night, the following remain active: – refrigerator, – heating system, – Wi-Fi router, – standby devices, – phone charging.

Periods of Absence

Aleksandra may leave the apartment for:

several hours

one night

a weekend

several days

Before leaving, she does not always change the settings of all household systems. Therefore, some energy consumption may continue even when the apartment is unoccupied.

Behavioural Variability

Aleksandra's energy consumption profile is highly variable.

The following should be taken into account:

- travel dates and duration may differ,
- departure and return times are irregular,
- the apartment may remain empty for several days,
- washing is often done shortly before or after a trip,
- standby consumption may continue during absence,
- heating-related energy consumption changes with weather conditions,
- meal times may vary,
- some days may show almost no active household consumption,
- days immediately before and after travel may show unusually high consumption,
- working from home may occasionally increase daytime electricity use.

Required Outcome

At the end of the challenge, the solution should include:

- | | |
|---|--|
| ☐ 1 a historical hourly energy consumption profile covering a minimum of 30 consecutive days, | ☐ 2 an energy consumption forecasting model, |
| ☐ 3 an hourly forecast for the next 24 hours, | ☐ 4 the ability to present forecasts for the next 3 and 7 days, |
| ☐ 5 an application presenting the results, | ☐ 6 identification of the hours with the highest expected demand, |
| ☐ 7 a brief explanation of changes in forecast energy consumption, | ☐ 8 a simulator showing the impact of a renewable energy investment on the household's energy balance and costs. |

Final Objective

The solution should make it possible to answer the following questions:

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- Q1** What energy demand can be expected in this household over the next 24 hours, 3 days, and 7 days?
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- Q2** During which hours will energy demand be highest?
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- Q3** What changes in household habits could be proposed to optimise the use of renewable energy and electricity from the grid?
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- Q4** How could the selected renewable energy investment affect the amount of electricity purchased from the grid, household costs, and the investment payback period?
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